

Hidden CGM Abnormalities & HbA1c-GMI Discordance in Controlled T2DM

Critical Appraisal of a Cross-Sectional Study at a Tertiary Center in North India

Department of Medicine, Santosh Medical College Ghaziabad

Cross-Sectional Observational Study

Hidden CGM Abnormalities and HbA1c-GMI Discordance in Apparently Controlled Type 2 Diabetes on Oral Antidiabetic Therapy

Abstract

Background: HbA1c remains the standard marker of glycemic control in type 2 diabetes mellitus (T2DM), but it does not capture hypoglycemia, time in range (TIR), or glycemic variability. Continuous glucose monitoring (CGM) can therefore uncover clinically important risk even when HbA1c appears acceptable.

Methods: We performed a cross-sectional analysis of an observational 14-day CGM dataset from adults with T2DM treated with oral antidiabetic drugs at a tertiary center in North India. The primary analysis included patients with HbA1c <8%. A CGM-unsafe phenotype was defined as any of TIR <70%, time below range (TBR) >4%, or coefficient of variation (CV) >36%. A sensitivity analysis used HbA1c <7%. Secondary analyses evaluated HbA1c-glucose management indicator (GMI) discordance in the full cohort of 107 patients. Mild discordance was defined as absolute HbA1c-GMI difference $\geq 0.50\%$ and major discordance as $\geq 1.00\%$.

Results: Of 107 patients, 63 had HbA1c <8% and formed the primary hidden-risk cohort. Their mean age was 53.50 ± 12.40 years, 46.0% were women, and mean HbA1c was $6.89 \pm 0.67\%$. Despite acceptable HbA1c, 47/63 patients (74.6%) had a CGM-unsafe phenotype. Low TIR <70% was present in 41/63 (65.1%), TBR >4% in 40/63 (63.5%), and CV >36% in 24/63 (38.1%). HbA1c was similar in CGM-safe and CGM-unsafe patients (6.94% vs 6.87%; $p=0.67$), whereas TIR (84.1% vs 61.9%), TBR (1.2% vs 11.9%), CV (25.6% vs 33.9%), SD (32.9 vs 44.8 mg/dL), and MODD (26.8 vs 35.6 mg/dL) differed significantly. Hidden risk was strongly associated with sulfonylurea (SU) exposure (96.7% vs 54.5%; $p<0.001$) and remained common with HbA1c <7% (24/30, 80.0%). In the full cohort, mean HbA1c was 7.68% and mean

Trial Identity and Citation

TRIAL NAME & CITATION

CGM-OAD India Trial: Hidden CGM Abnormalities and HbA1c-GMI Discordance in Apparently Controlled Type 2 Diabetes on Oral Antidiabetic Therapy

Citation: Tayal S et al

FUNDING & RESEARCH GROUPS

- Full Citation: Tayal S et al. Hidden CGM Abnormalities and HbA1c-GMI Discordance in Apparently Controlled Type 2 Diabetes on Oral Antidiabetic Therapy. Preprint 2026.
- Study Center: Santosh Medical College and Hospital, Ghaziabad, Uttar Pradesh, India.
- Funding Support: Investigator-initiated academic study; zero pharmaceutical/industry funding.

TRIAL REGISTRATION & SPONSORS

- Ethical Oversight: Approved by the Institutional Ethics Committee (IEC) of Santosh Hospital.
- Informed Consent: Written informed consent was obtained from all patients prior to CGM wear.
- Software & Registry: Analysis registry in Python 3.13 and R 4.2.2.



GLUCOSE STATISTICS AND TARGETS

3 April 2021 - 16 April 2021

% Time Sensor is Active

14 Days

88%

Ranges And Targets For	Type 1 or Type 2 Diabetes
Glucose Ranges	Targets % of Readings (Time/Day)
Target Range 70-180 mg/dL	Greater than 70% (16h 48min)
Below 70 mg/dL	Less than 4% (58min)
Below 54 mg/dL	Less than 1% (14min)
Below 180 mg/dL	Less than 25% (6h)
Below 250 mg/dL	Less than 5% (1h 12min)
Each 5% increase in time in range (70-180 mg/dL) is clinically beneficial.	

Average Glucose

156 mg/dL

Glucose Management Indicator (GMI)

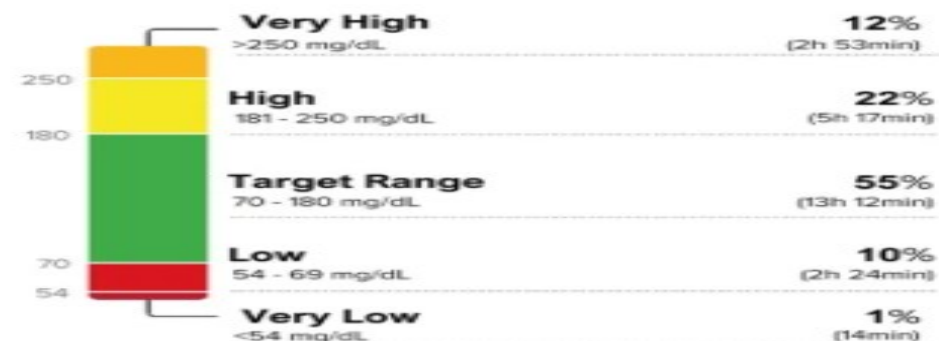
7.0% or 54 mmol/mol

Glucose Variability

46.2%

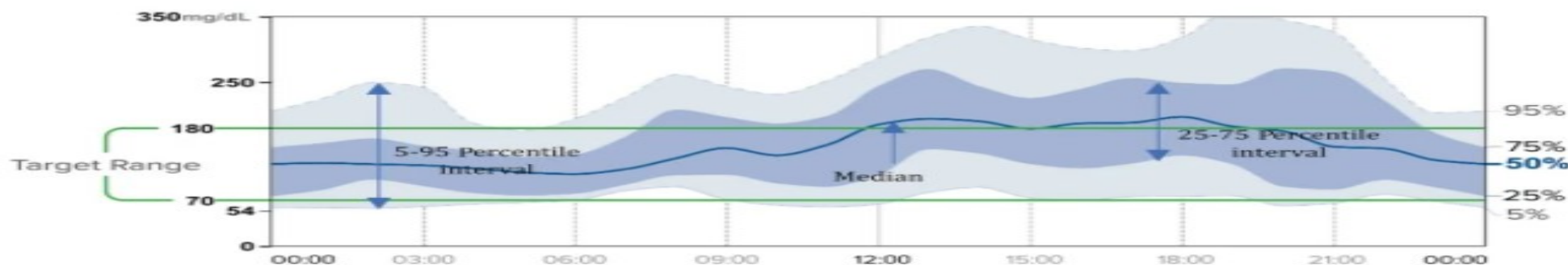
Defined as percent coefficient of variation (%CV); target $\leq 36\%$

TIME IN RANGES



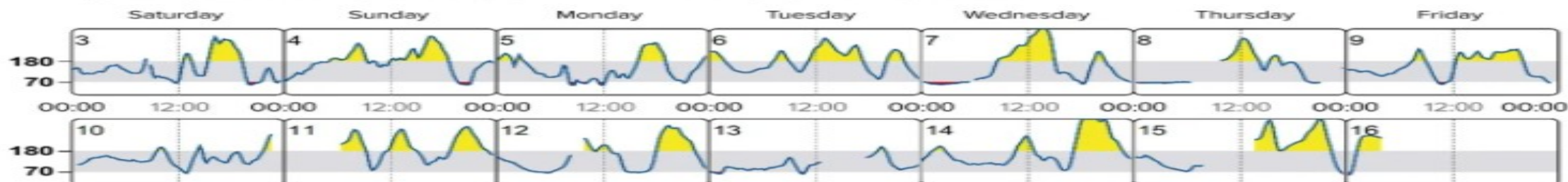
AMBULATORY GLUCOSE PROFILE (AGP)

AGP is a summary of glucose values from the report period, with median (50%) and other percentiles shown as if occurring in a single day.



DAILY GLUCOSE PROFILES

Each daily profile represents a midnight to midnight period with the date displayed in the top-left corner.



Executive Summary & Practice Bottom Line

- Clinical Question: Does acceptable HbA1c mask unsafe glucose patterns (hypoglycemia, high variability) and discordance with CGM-derived GMI in adults with T2DM on oral therapy?
- Population & Setting: 107 adults with T2DM on stable oral therapy at a tertiary hospital in Ghaziabad, North India.
- Key Efficacy Results: 74.6% of patients with HbA1c <8.0% had a CGM-unsafe phenotype (TIR <70%, TBR >4%, or CV >36%). Sulfonylurea (SU) exposure was strongly linked to unsafe patterns (96.7% vs 54.5%, $p<0.001$).
- GMI Discordance Results: Major HbA1c-GMI discordance ($\geq 1.0\%$) occurred in 41.1% of patients, all in the direction of HbA1c > GMI. This discordance was associated with lower eGFR, higher UACR, and SU exposure.
- Harms & Risks: Hypoglycemia (TBR >4%) was highly prevalent (63.5% in the <8% cohort, rising to 80.0% in the <7% cohort), representing a major hidden safety concern.

Practice Bottom Line: Consider targeted CGM in patients on sulfonylureas or with renal impairment, even if laboratory HbA1c appears controlled (<8.0% or <7.0%), to detect silent hypoglycemia and manage biomarker discordance.

Introduction: Continuous Glucose Monitoring (CGM)

HOW CGM WORKS

- Interstitial Measurement: A subcutaneous sensor measures glucose levels in the interstitial fluid every 5 to 15 minutes, generating 96 to 288 readings daily.
- 14-Day Sampling: Standard consensus recommends a 14-day wear duration, which captures approximately 90-95% of the variability of a 3-month HbA1c.
- Glycemic Dimensions: Captures three essential dimensions: average glucose, glycemic variability (instability), and time spent in hypo/hyperglycemic ranges.

THE CLINICAL RATIONALE

- The Fallacy of Average: Two patients with identical HbA1c can have completely different profiles: one stable in-range, the other swinging between severe lows and highs.
- Glycemic Excursions: Captures nocturnal events, postprandial spikes, and rapid rate-of-change fluctuations that capillary fingersticks (SMBG) almost always miss.

Key Advantage: CGM reveals the 24-hour glucose profile, overcoming the diagnostic blind spots of laboratory HbA1c.

Introduction: Consensus CGM Targets

TIME IN RANGE (TIR)

- Target Range: 70 to 180 mg/dL.
- Clinical Target: >70% of readings in target for most non-pregnant adults with T2DM.
- Clinical Value: Strongly correlated with a lower risk of long-term microvascular and macrovascular complications.

TIME BELOW RANGE (TBR)

- Hypoglycemia Threshold: <70 mg/dL (Level 1) and <54 mg/dL (Level 2).
- Clinical Target: <4% of readings <70 mg/dL, and <1% of readings <54 mg/dL.
- Clinical Value: Critical safety target to prevent acute neuroglycopenia, cognitive impairment, and cardiac arrhythmias.

GLYCEMIC VARIABILITY (CV)

- Metric: Coefficient of Variation (CV) calculated as standard deviation divided by mean glucose (expressed as a percentage).
- Clinical Target: $\leq 36\%$ (denotes stable glycemia; $>36\%$ indicates high variability).
- Clinical Value: Highly variable glucose is associated with oxidative stress and increased hypoglycemia risk.

Introduction: Understanding GMI and Discordance

GMI

Glucose Management Indicator

$\geq 0.5\%$

Mild HbA1c-GMI Discordance

$\geq 1.0\%$

Major HbA1c-GMI Discordance

14 Days

Minimum Sensor Wear Time

- GMI Formula: Estimated HbA1c based on average sensor glucose: $GMI (\%) = 3.318 + 0.00609 * \text{Mean Glucose (mg/dL)}$.
- Concept of Discordance: Laboratory HbA1c and CGM-derived GMI often differ. When $HbA1c > GMI$, it indicates that laboratory HbA1c may overestimate ambient glucose.
- Non-Glycemic Influences: Discordance is driven by red blood cell survival rates, iron deficiency, hemoglobinopathies, pregnancy, and chronic kidney disease (CKD).
- Clinical Trap: Over-relying on HbA1c when $HbA1c > GMI$ can lead to inappropriate treatment intensification, directly increasing the patient's risk of severe hypoglycemia.

Introduction: Why CGM Matters in Indian Practice

MEDICATION & AFFORDABILITY FACTORS

- **Sulfonylurea Reliance:** Sulfonylureas (SUs) remain widely used in India due to low cost, high potency, and wide availability in public and private clinics.
- **High Hypoglycemia Risk:** SUs stimulate insulin secretion independently of ambient glucose levels, significantly increasing the risk of hypoglycemia.
- **Glycemic Variability:** South Asian patients with T2DM often experience high glycemic excursions, postprandial spikes (high carbohydrate intake), and nocturnal instability.

CLINICAL MONITORING NEEDS

- **Silent Hypoglycemia:** Many patients do not experience classical autonomic warning symptoms of hypoglycemia, leaving them unaware of dangerously low values.
- **Consensus Guidance:** Recent Indian consensus statements recommend time-in-range (TIR) targets and expanding CGM access for patients on insulin or sulfonylureas.

Indian Perspective: In patients receiving sulfonylureas, CGM is vital to identify silent nocturnal lows that HbA1c alone cannot reflect.

STUDY OBJECTIVES & DESIGN

Clinical Questions, PICO Framework, and Appraisal Roadmap

Why This Paper Matters

Clinical problem: Clinical Problem: HbA1c is a mean marker and does not show whether a patient with apparently controlled T2DM has frequent, life-threatening silent hypoglycemia.

Evidence gap: Evidence Gap: Little data exists on the prevalence of hidden CGM abnormalities and HbA1c-GMI discordance under oral therapy in Indian settings, where SUs are common.

Research question: Research Question: How frequently do adults with T2DM on oral drugs and acceptable HbA1c exhibit unsafe CGM phenotypes and major HbA1c-GMI discordance in a tertiary North Indian hospital?

Hidden CGM Abnormalities and HbA1c-GMI Discordance in Apparently Controlled Type 2 Diabetes on Oral Antidiabetic Therapy

Abstract

Background: HbA1c remains the standard marker of glycemic control in type 2 diabetes mellitus (T2DM), but it does not capture hypoglycemia, time in range (TIR), or glycemic variability. Continuous glucose monitoring (CGM) can therefore uncover clinically important risk even when HbA1c appears acceptable.

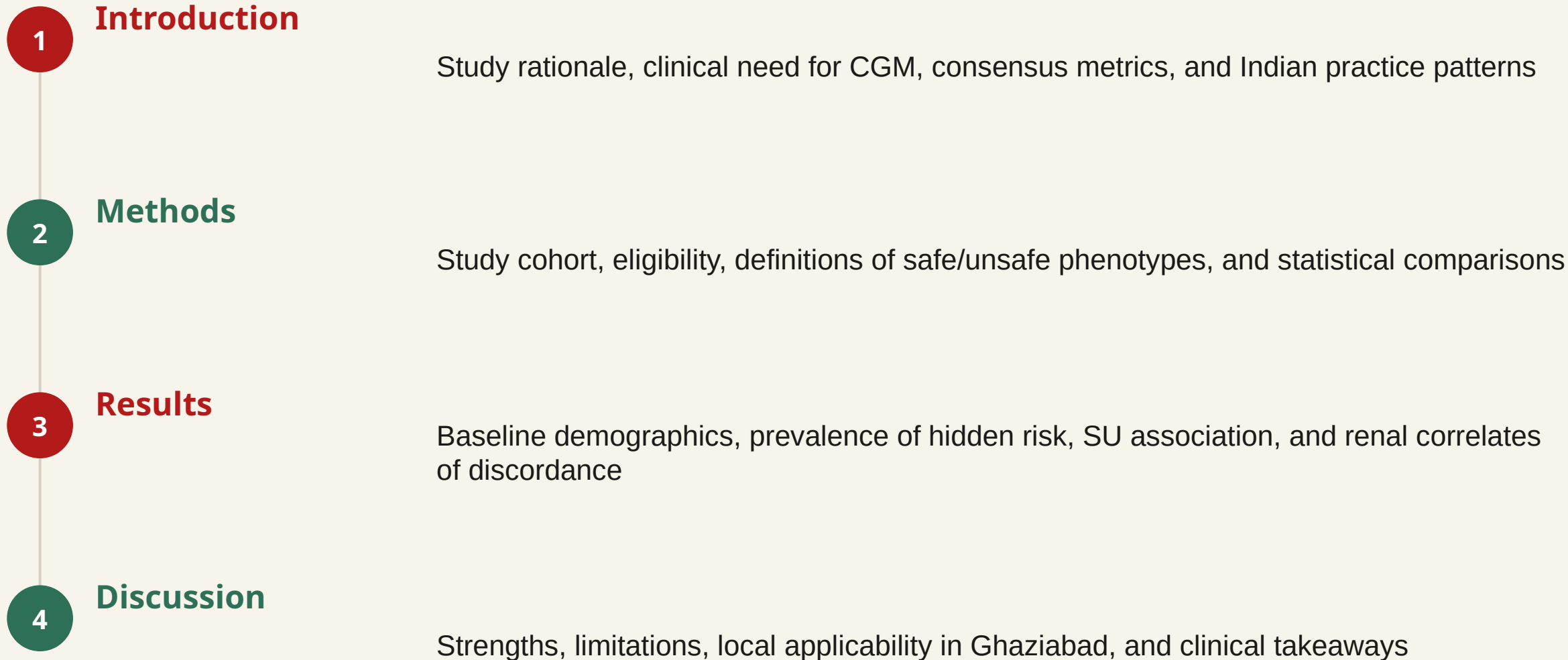
Methods: We performed a cross-sectional analysis of an observational 14-day CGM dataset from adults with T2DM treated with oral antidiabetic drugs at a tertiary center in North India. The primary analysis included patients with HbA1c <8%. A CGM-unsafe phenotype was defined as any of TIR <70%, time below range (TBR) >4%, or coefficient of variation (CV) >36%. A sensitivity analysis used HbA1c <7%. Secondary analyses evaluated HbA1c-glucose management indicator (GMI) discordance in the full cohort of 107 patients. Mild discordance was defined as absolute HbA1c-GMI difference $\geq 0.50\%$ and major discordance as $\geq 1.00\%$.

Results: Of 107 patients, 63 had HbA1c <8% and formed the primary hidden-risk cohort. Their mean age was 53.50 ± 12.40 years, 46.0% were women, and mean HbA1c was $6.89 \pm 0.67\%$. Despite acceptable HbA1c, 47/63 patients (74.6%) had a CGM-unsafe phenotype. Low TIR <70% was present in 41/63 (65.1%), TBR >4% in 40/63 (63.5%), and CV >36% in 24/63 (38.1%). HbA1c was similar in CGM-safe and CGM-unsafe patients (6.94% vs 6.87%; $p=0.67$), whereas TIR (84.1% vs 61.9%), TBR (1.2% vs 11.9%), CV (25.6% vs 33.9%), SD (32.9 vs 44.8 mg/dL), and MODD (26.8 vs 35.6 mg/dL) differed significantly. Hidden risk was strongly associated with sulfonylurea (SU) exposure (96.7% vs 54.5%; $p<0.001$) and remained common with HbA1c <7% (24/30, 80.0%). In the full cohort, mean HbA1c was 7.68% and mean

Clinical Question & PICO Framework

- P** — Adults aged 25 years or older with T2DM on stable oral antidiabetic therapy for 3+ months attending Santosh Hospital, Ghaziabad.
- I** — Continuous glucose monitoring (CGM) via a 14-day FreeStyle Libre sensor.
- C** — Stratification by HbA1c control (<8.0% vs ≥8.0%; <7.0% vs ≥7.0%) and oral therapy regimens (sulfonylurea-exposed vs non-exposed).
- O** — Prevalence of CGM-unsafe phenotype (TIR <70%, TBR >4%, or CV >36%) and HbA1c-GMI discordance (mild: ≥0.5%, major: ≥1.0%).

Structure of This Appraisal



STUDY METHODS

Participant Eligibility, Cohort Definitions, and Statistical Analysis

Study Design and Methodology

STUDY DESIGN & SETTING

- Study Type: Cross-sectional, retrospective analysis of an observational CGM registry.
- Study Center: Department of Medicine, Santosh Medical College and Hospital, Ghaziabad, Uttar Pradesh, India.
- Cohort Size: Full cohort of 107 patients analyzed for discordance; primary cohort of 63 patients with HbA1c <8.0% analyzed for hidden risk.

CGM SENSOR DETAILS

- Device Used: Abbott FreeStyle Libre 14-day flash glucose monitoring system.
- Sensor Wear: Participants wore the sensor for 14 days; metrics were extracted from stable 14-day glucose profiles.

Ethics and Design: Investigator-initiated, cross-sectional analysis of observational registry data approved by the Institutional Ethics Committee.

Eligibility Criteria

KEY INCLUSION CRITERIA

- Age Range: Adults aged 25 years or older with type 2 diabetes mellitus.
- Treatment Status: Stable oral antidiabetic drug (OAD) therapy for at least 3 months prior to enrollment.
- Sensor Wear: Agreement and ability to wear the CGM sensor continuously for 14 days.

KEY EXCLUSION CRITERIA

- Insulin Therapy: History of current or prior insulin use (to isolate OAD-specific risk).
- Severe Comorbidities: Severe renal impairment (eGFR <30 mL/min/1.73m²) or severe hepatic disease.
- Other Exclusions: Type 1 diabetes, pregnancy, breastfeeding, or use of medications known to disrupt glucose metabolism.

Cohort Selection: Designed to select stable, OAD-treated outpatient adults with type 2 diabetes.

Statistical Plan & Analysis

n = 107

Full Discordance Cohort

n = 63

HbA1c < 8.0% Subcohort

n = 30

HbA1c < 7.0% Sensitivity

p < 0.05

Statistical Significance

- Continuous comparisons: Compared CGM-safe and CGM-unsafe groups using Welch's t-test (since variances were unequal).
- Categorical comparisons: Compared percentages using Fisher's exact test (for small subgroup sizes) or Chi-square tests.
- Association testing: Analyzed the impact of sulfonylurea (SU) exposure on both hidden risk and GMI discordance.
- Sensitivity Analysis: Repeated the hidden risk analysis using a stricter target of HbA1c < 7.0% (n = 30).
- Software Tools: Statistical calculations performed in Python 3.13 (pandas, SciPy) and R 4.2.2.

STUDY RESULTS

Baseline Characteristics, Hidden Risk, SU Associations, and GMI Discordance

Participant Flow Diagram

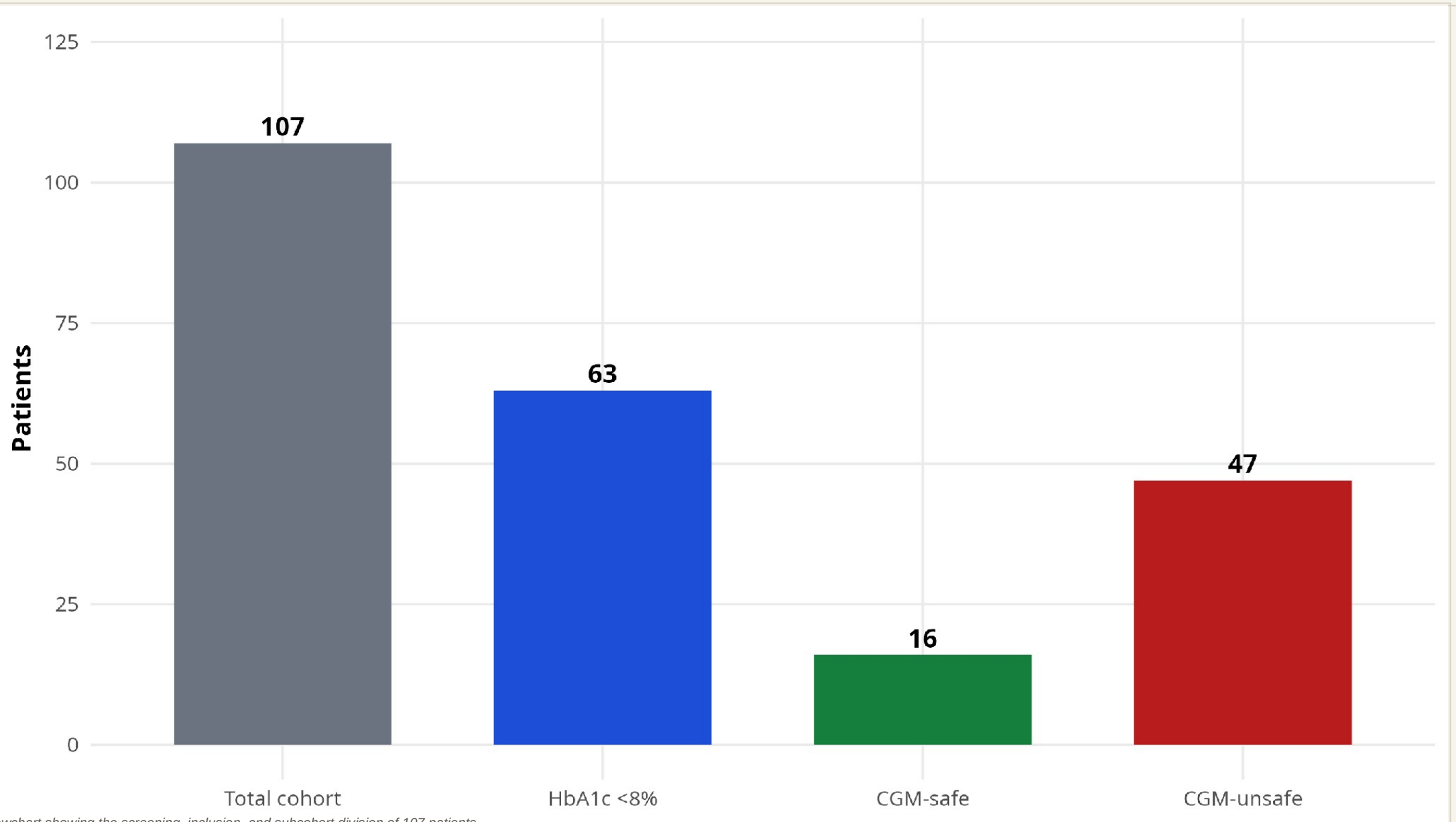


Figure 1: Flowchart showing the screening, inclusion, and subcohort division of 107 patients.

Table 1: Baseline Characteristics of Primary Cohort

Parameter	All Patients (n=63)	Regimen Distribution
Age (years)	53.51 ± 12.43	Metformin + DPP4i: 22 (34.9%)
Female sex, n (%)	29 (46.0%)	Metformin + DPP4i + SGLT2i: 11 (17.5%)
Duration of diabetes (years)	4.00 (2.00-7.50)	Metformin + SU: 17 (27.0%)
BMI (kg/m2)	24.74 ± 3.20	Metformin + SU + DPP4i: 13 (20.6%)
HbA1c (%)	6.89 ± 0.67	Total Sulfonylurea (SU) Exposed: 30 (47.6%)
GMI (%)	5.99 ± 0.89	Total Non-SU Exposed: 33 (52.4%)
eGFR (mL/min/1.73m2)	84.20 ± 31.69	Average glucose: 122.41 ± 26.35 mg/dL
UACR (mg/dL)	49.84 ± 28.55	Note: All patients had laboratory HbA1c < 8.0%

Baseline Characteristics Critique

- **Controlled Cohort:** The primary subcohort represents patients with apparently controlled glycemia (mean HbA1c $6.89\% \pm 0.67\%$).
- **Short Duration:** Median diabetes duration was 4.0 years, suggesting early-to-mid stage diabetes where beta-cell function is partially preserved.
- **Renal Profile:** Mean eGFR was 84.2 mL/min/1.73m², indicating mild renal impairment on average, with microalbuminuria present (mean UACR 49.8 mg/dL).
- **Treatment Split:** The cohort was balanced between SU-containing regimens (47.6%) and non-SU regimens (52.4%), allowing robust comparison.

Key Takeaway: *The cohort represents middle-aged patients with short-to-moderate diabetes duration. There is a high baseline use of sulfonylureas (47.6%), typical of Indian clinical practice.*

Table 2: Prevalence of Hidden CGM Risk (HbA1c < 8%)

CGM Phenotype / Target Component	Patients Meeting Criteria (n/63)	Percentage (%)
CGM-unsafe phenotype (Any of TIR <70%, TBR >4%, or CV >36%)	47	74.6%
Low Time in Range (TIR < 70%)	41	65.1%
Time Below Range (TBR > 4%) [Excess Hypoglycemia]	40	63.5%
High Glycemic Variability (CV > 36%)	24	38.1%
CGM-safe phenotype (TIR >=70%, TBR <=4%, and CV <=36%)	16	25.4%

Components of Hidden CGM Risk

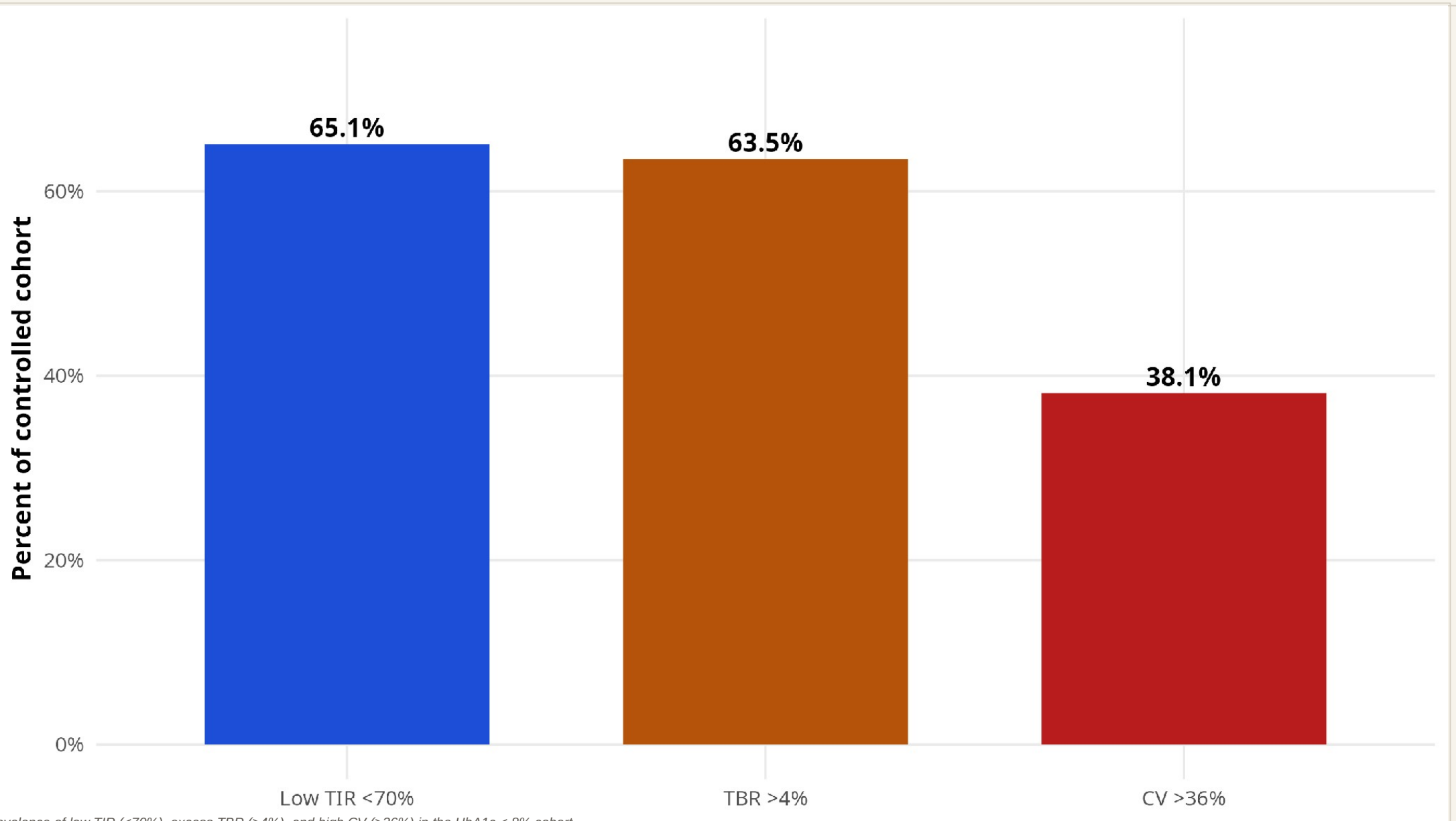


Figure 2: Prevalence of low TIR (<70%), excess TBR (>4%), and high CV (>36%) in the HbA1c < 8% cohort.

Patterns of Hidden CGM Abnormalities

DOMINANT ABNORMALITY PATTERNS

- Combined Defects: The commonest pattern was simultaneous low TIR, high TBR, and high CV, affecting 21 out of 63 patients (33.3%).
- TIR & TBR Joint Risk: Low TIR plus high TBR without high CV occurred in 13 out of 63 patients (20.6%).

ISOLATED DEFECTS

- Isolated Hypoglycemia: Isolated high TBR (with normal TIR and CV) was present in 5 out of 63 patients (7.9%).
- Isolated Low TIR: Isolated low TIR (with normal TBR and CV) was present in 5 out of 63 patients (7.9%).

Clinical Insight: Over 33% of patients had a combined defect of low TIR, excess TBR, and high CV, indicating severe overall glycemic instability.

Table 3: Glycemic Comparison of Safe vs Unsafe Phenotypes

Metric	CGM-Safe (n=16)	CGM-Unsafe (n=47)	Difference	P-Value
HbA1c (%)	6.94 ± 0.51	6.87 ± 0.71	-0.07%	0.67 (NS)
GMI (%)	6.21 ± 0.82	5.91 ± 0.91	-0.30%	0.23 (NS)
Avg Glucose (mg/dL)	128.2 ± 24.5	120.4 ± 26.9	-7.8 mg/dL	0.30 (NS)
Time in Range (TIR, %)	84.1% ± 10.1%	61.9% ± 8.0%	-22.3%	<0.001
Time Below Range (TBR, %)	1.2% ± 1.3%	11.9% ± 9.6%	+10.7%	<0.001
Time Above Range (TAR, %)	14.8% ± 9.6%	25.6% ± 9.0%	+10.8%	<0.001
Coefficient of Variation (CV, %)	25.6% ± 5.6%	33.9% ± 8.8%	+8.3%	<0.001
Standard Deviation (SD, mg/dL)	32.9 ± 9.8	44.8 ± 14.8	+11.9	<0.001

Comparison of Glycemic Metrics (Safe vs Unsafe)

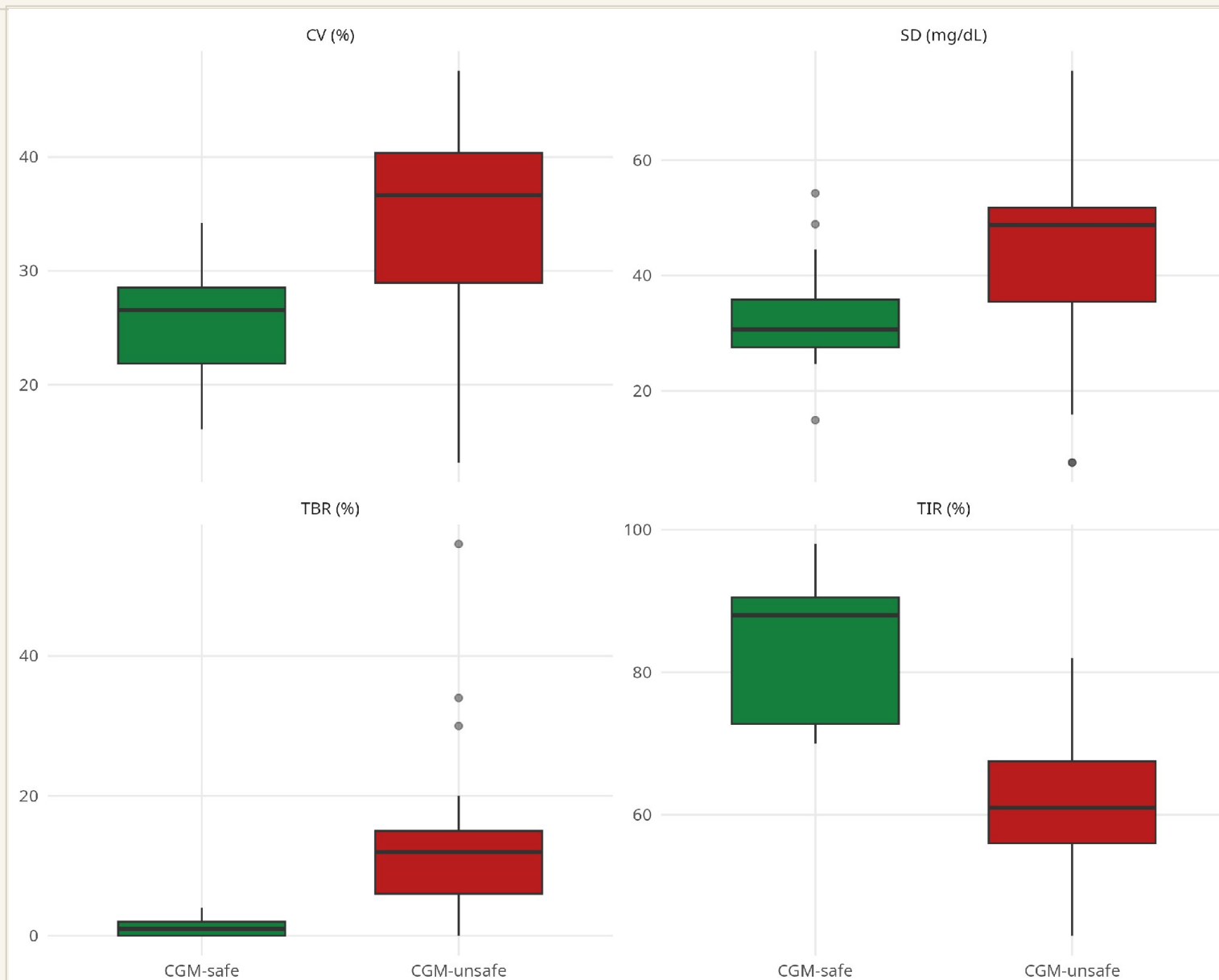


Figure 5: Boxplots showing that HbA1c overlaps completely between CGM-safe and CGM-unsafe groups, whereas TIR and TBR are widely separated.

Appraisal of Glycemic Markers Failure

- Biomarker Blindness: Mean HbA1c was numerically lower in the unsafe group (6.87% vs 6.94%), yet this group spent a median of ~2.8 hours per day in hypoglycemia.
- Variability Divergence: Standard Deviation (SD) and Coefficient of Variation (CV) were both significantly higher in the unsafe group, showing unstable glycemic excursions.
- Daily Differences: Mean of Daily Differences (MODD) was significantly higher in the unsafe group (35.6 vs 26.8 mg/dL; $p=0.006$), denoting high day-to-day instability.
- Mean Glucose Trap: Average sensor glucose (120 mg/dL vs 128 mg/dL) did not differ significantly, illustrating that mean-based metrics fail to reflect risk.

The Fallacy of HbA1c: Laboratory HbA1c was completely blind to a 10-fold difference in hypoglycemia exposure (TBR 11.9% vs 1.2%) and a 22% reduction in time-in-range.

Hidden CGM Risk by Medication Regimen

50.0%

Met + DPP4i (11/22)

63.6%

Met + DPP4i + SGLT2i (7/11)

100.0%

Met + SU (17/17)

92.3%

Met + SU + DPP4i (12/13)

- Regimen differences: The prevalence of CGM-unsafe phenotypes differed significantly across the four oral treatment regimens ($p < 0.01$).
- DPP4i Safety: Metformin + DPP4i had the lowest hidden risk (50.0%), reflecting the low intrinsic hypoglycemia risk of DPP-4 inhibitors.
- Triple OAD: Metformin + DPP4i + SGLT2i had intermediate risk (63.6%), likely reflecting baseline clinical differences.
- Sulfonylurea Risk: Both regimens containing sulfonylureas had near-universal unsafe phenotypes: 100% of Met+SU and 92.3% of Met+SU+DPP4i patients were unsafe.

Unsafe CGM Phenotype by Medication Regimen

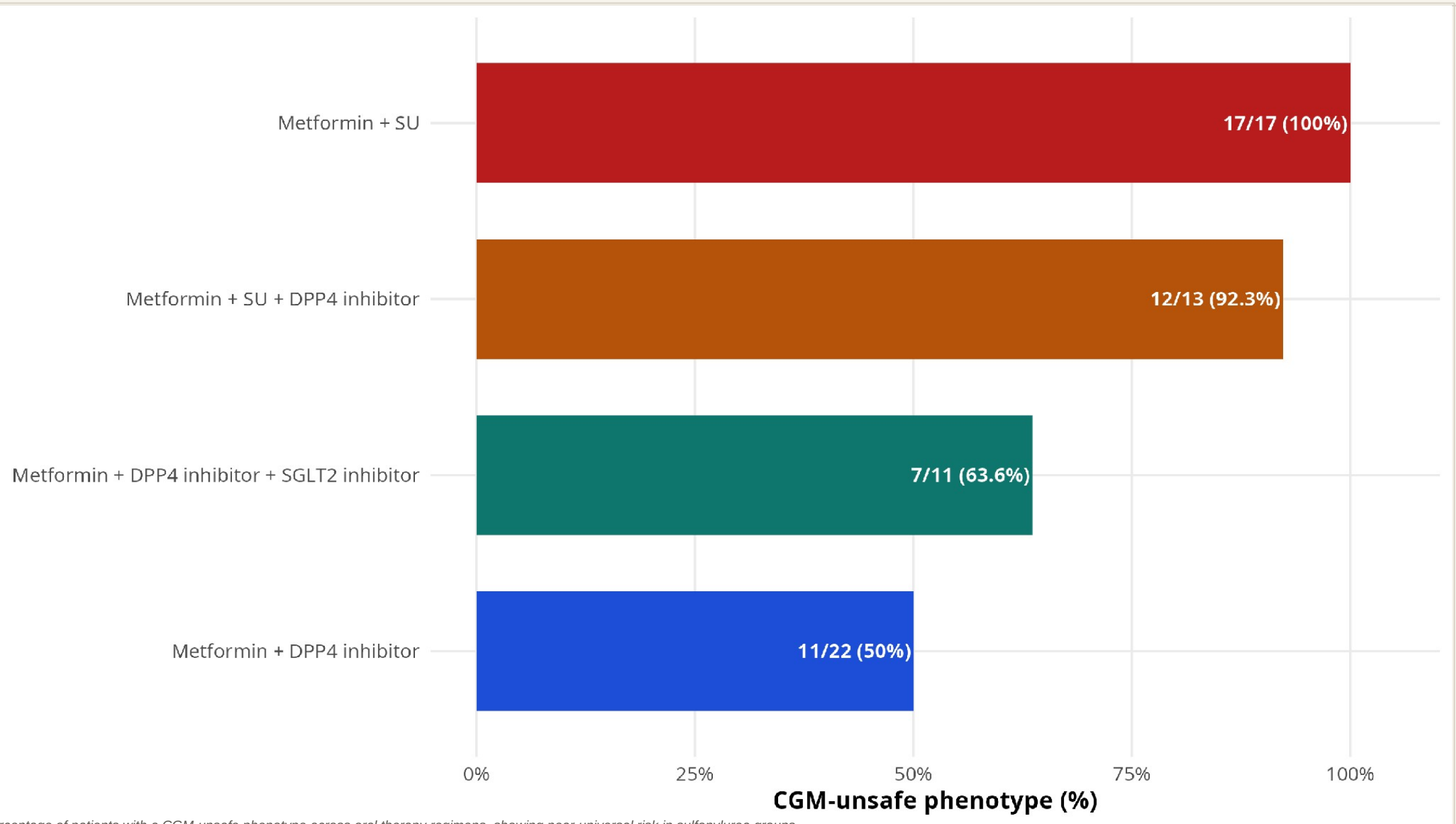


Figure 3: Percentage of patients with a CGM-unsafe phenotype across oral therapy regimens, showing near-universal risk in sulfonylurea groups.

Sulfonylurea Exposure and Glycemic Safety

SULFONYLUREA EXPOSED (n=30)

- Universal Unsafe Phenotype: 29 out of 30 SU-exposed patients (96.7%) met the criteria for a CGM-unsafe phenotype.
- High Hypoglycemia Rates: Excess TBR was almost universal in the SU group, reflecting drug-induced insulin release during fasting or post-absorptive states.

NON-SULFONYLUREA REGIMENS (n=33)

- Controlled but Unsafe: 18 out of 33 non-SU patients (54.5%) were classified as unsafe, which is still a substantial proportion.
- Regimen Choice: Demonstrates that even when HbA1c is successfully controlled, SU-based OAD therapy is associated with high glycemic risk.

Medication Signal: Sulfonylurea exposure was the single strongest clinical correlate of hidden CGM abnormalities (Fisher exact $p < 0.001$).

Sensitivity Analysis: Stricter Control (HbA1c < 7%)

80.0%

CGM-Unsafe (24/30)

66.7%

Low TIR < 70% (20/30)

80.0%

Excess TBR > 4% (24/30)

43.3%

High CV > 36% (13/30)

- Threshold Robustness: The hidden-risk signal remained extremely strong when applying a stricter target of HbA1c < 7.0% (n = 30).
- Higher Hypoglycemia: In this tighter control group, the percentage of patients with excess hypoglycemia (TBR >4%) actually rose to 80.0%.
- Persistent Instability: Almost half of the patients (43.3%) continued to show high glycemic variability (CV >36%).
- Conclusion: The discordance between HbA1c and CGM safety targets is not an artifact of a loose HbA1c cutoff of 8.0%; it is even more prominent at normal clinical targets (<7.0%).

HbA1c-GMI Discordance in the Full Cohort

1.02%

Mean HbA1c-GMI Diff

74.8%

Mild Discordance ($\geq 0.5\%$)

41.1%

Major Discordance ($\geq 1.0\%$)

100.0%

Major in Direction HbA1c > GMI

- Cohort Mean: In the full cohort of 107 patients, mean laboratory HbA1c was 7.68% and mean CGM-derived GMI was 6.66%.
- Mild Discordance: Mild discordance (absolute difference $\geq 0.50\%$) was present in 80/107 patients (74.8%), with 98.8% of these having HbA1c > GMI.
- Major Discordance: Major discordance (absolute difference $\geq 1.00\%$) occurred in 44/107 patients (41.1%).
- Unidirectional Trend: All 44 patients with major discordance had HbA1c > GMI, indicating a highly systematic biological bias.

HbA1c vs GMI Discordance Scatter Plot

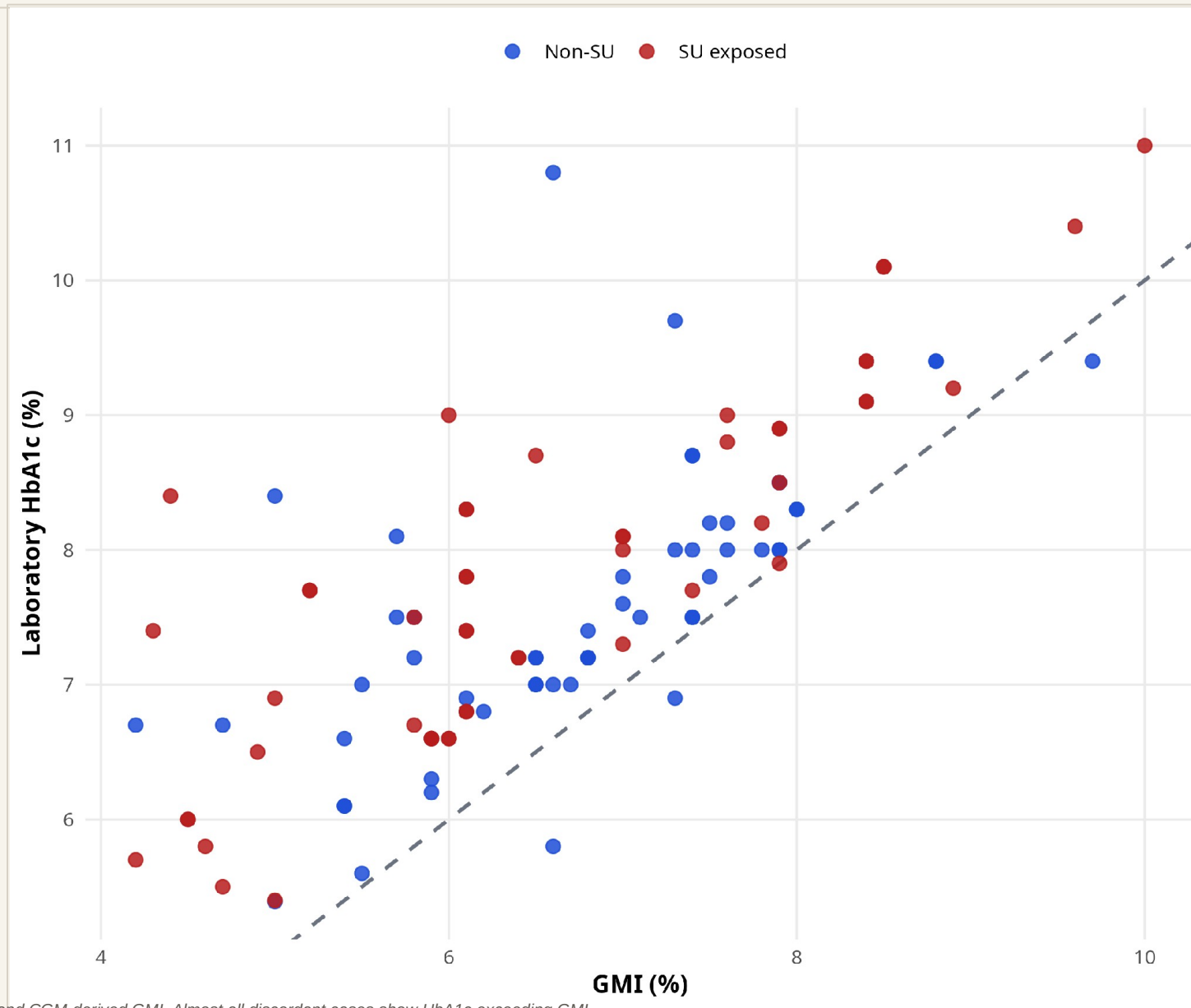


Figure 4: Relationship between laboratory HbA1c and CGM-derived GMI. Almost all discordant cases show HbA1c exceeding GMI.

Table 4: Clinical Correlates of Major GMI Discordance

Metric	Non-Major Diff <1% (n=63)	Major Diff >=1% (n=44)	Difference	P-Value
HbA1c (%)	7.43 ± 1.07	8.04 ± 1.27	+0.61%	0.01
GMI (%)	6.96 ± 1.10	6.23 ± 1.40	-0.73%	0.005
Avg Glucose (mg/dL)	150.5 ± 34.6	134.7 ± 52.3	-15.9 mg/dL	0.08
Time Below Range (TBR, %)	5.7% ± 6.2%	10.5% ± 10.2%	+4.8%	0.007
eGFR (mL/min/1.73m2)	97.7 ± 26.5	68.5 ± 31.7	-29.2 mL/min	<0.001
UACR (mg/dL)	39.6 ± 19.7	63.5 ± 34.4	+23.9 mg/dL	<0.001

Critique of Major GMI Discordance

- Renal Association: Mean eGFR was 68.5 in the major discordance group vs. 97.7 in the non-major group ($p < 0.001$), showing a strong renal link.
- Increased Hypoglycemia: Patients with major discordance had significantly higher TBR (10.5% vs 5.7%; $p=0.007$) and lower GMI, despite a higher HbA1c.
- Medication Link: Major discordance was strongly associated with sulfonylurea exposure (58.5% vs 24.1%; $p<0.001$), compounding hypoglycemia risk.
- Therapeutic Risk: If a clinician intensifies therapy based on the high HbA1c (8.04% vs 7.43%), they will worsen hypoglycemia in patients who already spend 10.5% of their time below range.

Renal Biomarker Signal: Lower eGFR and higher microalbuminuria are strongly associated with major discordance. In renal impairment, HbA1c systematically overestimates ambient glucose, creating a therapeutic trap.

STUDY APPRAISAL & DISCUSSION

Strengths, Limitations, Risk of Bias, GRADE, and Local Applicability

Results Critical Appraisal

STRENGTHS OF FINDINGS

- Patient-Level Data: Relies on continuous 14-day sensor data rather than intermittent capillary blood checks.
- Target Alignment: Uses internationally recognized consensus thresholds (TIR <70%, TBR >4%, CV >36%) to define risk.
- Clinical Relevance: Explores both OAD regimen patterns and renal function, linking biomarkers to clinical therapy.

LIMITATIONS & THREATS

- Sample Size: Moderate sample size (n=107) from a single center, limiting broad generalizability.
- Cross-Sectional Design: Prevents making causal assertions about the relationship between renal function, OADs, and discordance.

Internal Validity: Results are highly consistent across both subcohort thresholds (<8% and <7%) and show strong medication and biomarker signals.

Observational Study Risk of Bias

SOURCES OF BIAS

- Selection Bias: Single-center recruitment from a tertiary hospital may overrepresent patients with higher complication rates or medication burdens.
- Missing Biomarkers: Lacked hemoglobin level, iron status, or direct measures of red blood cell survival, which are key biological drivers of discordance.
- Reverse Causality: Cannot determine whether renal impairment directly caused HbA1c-GMI discordance or if they are linked by a common pathway.

MITIGATION & VALIDITY

- Standardized Testing: Re-analyzed de-identified raw datasets using standardized consensus definitions, reducing reporting bias.
- Robust Sensitivity: Sensitivity analysis with HbA1c <7.0% confirmed the robustness of the primary hidden-risk prevalence.

Overall Bias Assessment: Moderate risk of bias due to lack of longitudinal follow-up and unmeasured confounders, typical of cross-sectional designs.

GRADE Certainty of Evidence

GRADE FINAL CERTAINTY

MODERATE CERTAINTY

We have moderate confidence in the effect estimate. The true effect is likely to be close to the estimate, but there is a possibility that it is substantially different.

**Outcome: Hidden CGM Risk & HbA1c-GMI
Discordance in OAD-treated T2DM**

Risk of Bias: MINIMAL (-0)

No downgrade. Standardized consensus criteria applied; de-identified analytical registry data.

Indirectness: MINIMAL (-0)

No downgrade. Direct measurement of clinical metrics (TIR, TBR, GMI) using standard devices in a representative outpatient cohort.

Inconsistency: MINIMAL (-0)

No downgrade. Consistency in SU and renal risk signals across multiple subgroups and sensitivity analyses.

Imprecision: SERIOUS (-1)

Downgraded by 1 level. Single-center study with a moderate sample size (n = 107). Requires validation in larger multicenter cohorts.

Chronology of Key CGM and Outcome Trials

Study (Year)	Study Type & N	Glycemic Biomarkers	Key Findings & Practice Insights
Beck et al. (2017)	Observational (n=200)	HbA1c vs. CGM mean	Coined 'Fallacy of Average'. Showed HbA1c fails to distinguish stable from highly variable glucose profiles.
Mohan et al. (2023)	Indian Consensus	TIR & CGM targets	Established South Asian targets (TIR >70%, TBR <4%). Recommended CGM to optimize OAD and insulin therapy.
Fang et al. (2023)	NHANES Cohort (n=6,458)	HbA1c vs. GMI	Evaluated GMI performance in T2DM. Found substantial discordance, showing HbA1c overestimates GMI in non-White races.
Singh et al. (2026)	Cross-Sectional (n=107)	TIR, TBR, HbA1c, GMI	Tertiary North Indian center. Found 74.6% hidden risk in 'controlled' T2DM. Linked discordance to eGFR, UACR, and sulfonylureas.

Current Guideline Consensus on CGM Targets

- ADA Standards of Care (2026): Recommends CGM-derived TIR and TBR alongside HbA1c as standard of care. Recommends GMI interpretation to evaluate biomarker discordance.
- RSSDI / Indian Consensus (2023-2024): Emphasizes TIR target >70% and TBR <4% for South Asian patients. Recommends retrospective CGM to optimize therapy in patients on SUs or insulin.
- AACE Guidelines (2025): Strongly supports CGM use in all patients with T2DM on therapies with high hypoglycemia risk (specifically sulfonylureas and insulin).
- Consensus Verdict: HbA1c remains the primary tool for long-term epidemiological tracking, but CGM metrics (TIR, TBR, CV) are the gold standard for clinical safety and therapeutic titration.

CLINICAL APPLICABILITY & TAKEAWAYS

Ghaziabad Hospital Setting, Ward Protocols, and Discussion Questions

Local Applicability: North Indian Patients

RELEVANCE & OPPORTUNITIES

- Sulfonylurea Prevalence: SUs (glimepiride, gliclazide) are widely prescribed in North Indian medical colleges due to low cost and availability under state health schemes.
- Renal Comorbidities: CKD is highly prevalent among diabetes outpatients in Ghaziabad, making HbA1c-GMI discordance a frequent clinical issue.
- Safety Optimization: Identifying patients with high TBR (>4%) allows clinicians to safely de-escalate SUs and switch to DPP4i or SGLT2i (if renal function allows).

BARRIERS & CHALLENGES

- Cost Constraints: Personal real-time CGM (e.g. Libre sensor) is expensive (~3000-4000 INR per 14 days) and not reimbursable for most patients.
- Professional CGM: A clinic-owned retrospective CGM reader can be shared across patients, reducing individual cost and providing actionable 14-day snapshots.

Local Clinical Context: In Ghaziabad, high SU use makes silent hypoglycemia a major risk. Targeted professional/retrospective CGM offers a cost-effective safety net.

Clinical Takeaways for OPD, Wards, & ICU/HDU

OUTPATIENT CLINICS (OPD)

- Screen SU Patients: Perform a targeted 14-day CGM in patients on sulfonylureas with 'acceptable' HbA1c to screen for silent hypoglycemia.
- De-escalate SUs: If TBR >4%, down-titrate or stop the sulfonylurea, replacing it with a DPP-4 inhibitor or other low-risk agent.
- Renal Check: In patients with eGFR <60 or microalbuminuria, suspect HbA1c-GMI discordance and avoid aggressive titration based on HbA1c.

INPATIENT WARDS

- Hold SUs on Admission: Always hold sulfonylureas during acute admission for infection, AKI, or dehydration to prevent severe hypoglycemia.
- Transition Targets: Use sliding-scale regular insulin or low-risk oral agents for inpatient glycemic control.
- Discharge Plan: Do not discharge patients on high doses of sulfonylureas if they experienced in-hospital hypoglycemia or have renal impairment.

ICU & HIGH DEPENDENCY (HDU)

- Target MAP & Perfusion: Maintain MAP ≥ 65 mmHg and avoid tight glucose control (target 140-180 mg/dL) in critically ill patients.
- Avoid SUs: Sulfonylureas have no role in the ICU/HDU due to long half-life, unpredictability, and risk of severe hypoglycemia.
- Renal Monitoring: Monitor creatinine and urine output hourly; adjust insulin infusion rates dynamically based on renal clearance.

Journal Club Discussion Questions

SULFONYLUREA DE-ESCALATION

1. In our outpatient clinics in Ghaziabad, how can we establish a protocol to safely de-escalate sulfonylureas in patients who appear controlled by HbA1c but have high hypoglycemia risk?

BIOMARKER DISCORDANCE

2. How should we educate our residents and fellows on the discrepancy between laboratory HbA1c and CGM-derived GMI, particularly in patients with chronic kidney disease?

CGM IMPLEMENTATION

3. What are the most cost-effective ways to implement professional (retrospective) CGM in a tertiary hospital setting like Santosh Hospital to overcome financial barriers?

GUIDELINE MODIFICATION

4. Should local clinical protocols in Indian medical colleges shift from HbA1c-only targets to incorporating time-in-range (TIR) and time-below-range (TBR) in routine audits?

References

- Danne T, Nimri R, Battelino T, et al. International consensus on use of continuous glucose monitoring. *Diabetes Care*. 2017;40(12):1631-1640. doi:10.2337/dc17-1600
- Battelino T, Danne T, Bergenstal RM, et al. Clinical targets for continuous glucose monitoring data interpretation: recommendations from the international consensus on time in range. *Diabetes Care*. 2019;42(8):1593-1603. doi:10.2337/dci19-0028
- Beck RW, Connor CG, Mullen DM, et al. The fallacy of average: how using HbA1c alone to assess glycemic control can be misleading. *Diabetes Care*. 2017;40(8):994-999. doi:10.2337/dc17-0636
- Mohan V, Joshi S, Mithal A, et al. Expert consensus recommendations on time in range for monitoring glucose levels in people with diabetes: an Indian perspective. *Diabetes Ther*. 2023;14(2):237-249. doi:10.1007/s13300-022-01355-4
- Fang M, Wang D, Rooney MR, et al. Performance of the glucose management indicator in type 2 diabetes. *Clin Chem*. 2023;69(4):422-428. doi:10.1093/clinchem/hvac210